

Setup Guide: Payload Transport

Time Allocation: 90 Minutes

Challenge Overview

The robot must carry a 500-gram weight across a fragile cardboard bridge without causing structural collapse.

Environment & Arena Setup

- **1. Arena Construction:** Set up a flat, stable testing surface and integrate a standardized cardboard bridge spanning a gap of approximately 30 to 40 centimeters. Ensure the bridge is constructed from a single layer of corrugated cardboard to maintain a consistent fragility level across all teams.
- **2. Payload Preparation:** Provide standardized 500-gram weights (such as weighted blocks or sandbags) that can be easily attached to the student robots.
- **3. Safety and Boundaries:** Use painter's tape to outline the driving lanes leading up to and exiting the bridge to keep robots aligned with the crossing path.
- **4. Inspection Area:** Create a designated testing station where instructors can weigh and inspect the student robots and payloads prior to their official run.